Total No. of Questions : 8]

SEAT No. :

PB-3851

[Total No. of Pages : 2

[6262]-113

T.E. (Electronics and Computer Engg.)

SENSORS AND APPLICATIONS

(2019 Pattern) (Semester - I) (310345D) (Elective-I)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q. 7or Q8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Assume Suitable data if necessary.***
4) ***Use of non-programmable scientific calculator is allowed.***

Q1) a) Explain with neat sketches the working principle of **[8]**

i) Doppler type ultrasonic flow meter and

ii) Transit time ultrasonic flow meter.

b) Describe with neat sketch electromagnetic flow meter. **[9]**

OR

Q2) a) Explain working principle of hydrostatic pressure type of level measurement technique. **[8]**

b) Explain with a neat sketch working principle of rotameter. Can a rotameter be used in a horizontal pipe line? If not, explain why? **[9]**

Q3) a) Explain with neat sketch working principle of thermal accelerometer. State applications of accelerometers. **[8]**

b) Explain working principle of Geiger Muller counter used for detection of nuclear radiation. **[9]**

OR

Q4) a) Explain the process of charge transfer in CMOS image sensors. **[8]**

b) Explain with neat sketch working principle of incremental optical encoder.

P.T.O.

what is the use of index pulse in incremental encoder? Explain how an incremental encoder is used to sense direction of rotation of shaft. **[9]**

Q5) a) Explain working principle of PZT actuators. State its applications. **[9]**

b) Explain magneto-transistor and magneto-resistive elements (MRE). **[9]**

OR

Q6) a) Write a short note on surface micromachining for MEMS devices. **[9]**

b) Explain with neat block diagram the concept of SMART sensor system. **[9]**

Q7) a) Draw and explain the symbols of following pneumatic valves. **[6]**

i) 5X2 valve

ii) 4X2 valve

iii) 3X2 valve

b) Explain how a solenoid is used as an actuator. **[6]**

c) Draw control valve characteristics and explain the terms **[6]**

i) Quick Opening

ii) Linear and

iii) Equal Percentage.

OR

Q8) a) A 5V control signal is to be used to turn ON and OFF a pump operating on 230VAC. Explain a relay driver circuit which can be used for this application. **[6]**

b) Explain control of single acting cylinder using an appropriate directional control valve. **[6]**

c) Explain how actuators are classified. Explain any one type of actuators. **[6]**

Total No. of Questions : 8]

SEAT No. :

P7610

[Total No. of Pages : 2

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T.E. (Electronics & Computer Engineering)
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(2019 Pattern) (Semester - I) (Elective - I) (310345 D)

Time : 2 ½ Hours] *[Max. Marks : 70*

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4) *Use of non-progranunable scientific calculator is allowed.*

Q1) a) Explain working principle of hydrostatic pressure type of level measurement technique. **[8]**

b) Explain with neat sketch working principal of following differential head type of flow sensors- **[9]**

i) Orifice plate

ii) Venturi Tube

OR

Q2) a) Explain with neat diagram working principal of nuclear level gauge. **[8]**

b) Explain with neat sketch working principal of pitot static tube used for the measurement of airspeed aboard an aircraft. **[9]**

Q3) a) Draw a neat sketch of capacitive accelerometer. Explain working principle of capacitive accelerometer. **[6]**

b) Explain with neat sketch working principle of incremental optical encoder. **[6]**

c) Explain working principal of optical proximity sensors. **[6]**

OR

Q4) a) Explain working principal of LVDT as a displacement sensor. **[6]**

b) Explain weight measurement technique using load cell and strain gauges. **[6]**

c) Write a short note on CMOS image sensors. **[6]**

P.T.O.

Q5) a) Explain working principle of PZT actuators. State its applications. **[8]**

b) Explain Hall effect type magnetic field sensors. State the names of materials used for Hall effect devices. **[9]**

OR

Q6) a) Explain with neat block diagram the concept of SMART sensor system. [8]

b) Explain with neat sketch bulk micro-machined absolute pressure sensor. [9]

Q7) a) Draw a pneumatic circuit symbol for a poppet valve. Explain pneumatic lift system using a poppet valves and single acting cylinder. **[6]**

b) Explain principle of operation of stepper motor. **[6]**

c) Draw control valve characteristics and explain the terms. **[6]**

i) Quick Opening

ii) Linear and

iii) Equal Percentage.

OR

Q8) a) Draw and explain the symbols of following pneumatic valves. **[6]**

i) 5×2 valve

ii) 4×2 valve

iii) 3×2 valve

b) Explain control of single acting cylinder using directional control valve. **[6]**

c) Explain how actuators are classified. Explain any one type of actuators. **[6]**

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P7610

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